

A "Guess the Hidden Number" circuit with a 6-bit address bus, 4-bit data bus, 1 kHz clock, 2114 RAM, 74LS73, 74LS00, 74LS244, 74LS85, 74LS93, switches, push-buttons, and LEDs using accessible tools and parts that are in-stock.

Build of Materials

Component	Quantity	Description
2114	1	1K x 4-bit static RAM IC (18-pin DIP)
LM555	1	555 Timer IC for 1 kHz clock (8-pin DIP)
74LS73	1	Dual JK flip-flop for debouncing (14-pin DIP)
74LS00	1	Quad 2-input NAND gate for control logic (14-pin DIP)
74LS244	1	Octal buffer/line driver for output LEDs (20-pin DIP)
74LS85	1	4-bit magnitude comparator for game logic (16-pin DIP)
74LS93	1	4-bit binary counter for guess tracking (14-pin DIP)
i8226	1	4-bit bus driver for data bus (16-pin DIP)
SPST Switch	11	6 for address (S1-S6), 4 for data (S7-S10), 1 for read/write (S11)
SPST Push-Button	2	LOAD button (S12), Reset button (S13, optional)
LED	15	4 for input (L1-L4), 4 for output (L5-L8), 3 for feedback (L9: red, L10: yellow, L11: green), 4 for counter (L12-L15)
Resistor (10 kΩ)	22	11 pull-up, 11 pull-down for switches
Resistor (330 Ω)	15	Current-limiting for LEDs
Resistor (10 kΩ)	1	LM555 timing resistor (R1)
Resistor (68 kΩ)	1	LM555 timing resistor (R2)
Capacitor (0.1 μF)	8	1 for LM555 timing (C8), 7 for decoupling (C1-C7, one per IC)
Power Supply	1	5V DC regulated supply

Schematic Description

Here, I describe the layout and key connections, refining the text-based schematic. The circuit is organized into functional blocks for clarity.

1. Power and Ground

- VCC (5V): Connects to all IC VCC pins (2114 pin 18, LM555 pin 8, 74LS73 pin 14, 74LS00 pin 14, 74LS244 pin 20, 74LS85 pin 16, 74LS93 pin 5).
- GND: Connects to all IC GND pins (2114 pin 9, LM555 pin 1, 74LS73 pin 7, 74LS00 pin 7, 74LS244 pin 10, 74LS85 pin 8, 74LS93 pin 10), LED cathodes, and switch pull-downs.
- Decoupling Capacitors (C1-C7, 0.1 μF):
 - Place one near each IC's VCC pin, connected between VCC and GND.
 - Example: C1 near 2114 (pin 18 to pin 9), C2 near LM555, etc.

2. Clock (LM555, 8-pin DIP)

- Purpose: 1 kHz clock for 74LS73 synchronization.
- Connections:

LM555:

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Pin 1 (GND) --- GND
Pin 2 (TRIG) --- Pin 6 (THR) & C8 [0.1μF] --- GND
Pin 3 (OUT) --- 74LS73 CLK1 (pin 1)
Pin 4 (RESET) --- VCC
Pin 5 (CTRL) --- [0.1μF] --- GND (optional)
Pin 6 (THR) --- Pin 2 & R1 [10kΩ] --- Pin 7
Pin 7 (DISCH) --- R1 [10kΩ] --- Pin 6 & R2 [68kΩ] --- VCC
Pin 8 (VCC) --- VCC & C2 [0.1μF] --- GND
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- Layout: Place LM555 top-left, with R1, R2, C8 nearby. Route OUT to 74LS73.

3. Control Logic (74LS73, 74LS00)

- 74LS73 (14-pin DIP, Dual JK Flip-Flop):

- Purpose: Debounces LOAD button (S12), generates pulse.
- Connections:

Pin 1 (CLK1) --- S12 (LOAD, 10kΩ pull-up to VCC, 10kΩ pull-down to GND)
Pin 2 (J1) --- VCC
Pin 3 (K1) --- GND
Pin 4 (CLR1) --- VCC
Pin 5 (Q1) --- 74LS00 pin 1 $\overline{CS}$, pin 4 $\overline{WE}$, 74LS93 CKA (pin 14), 74LS00 pin 9 (reset)
Pin 6 ($\overline{Q1}$) --- Unconnected
Pin 7 (GND) --- GND
Pin 8-13 (Second FF) --- Unconnected or CLK2 to LM555 OUT (pin 3), J2 to VCC, K2 to GND, CLR2 to VCC, Q2 to 74LS00 pin 10
Pin 14 (VCC) --- VCC & C3 [0.1μF] --- GND

- 74LS00 (14-pin DIP, Quad NAND):

- Purpose: Generates $\overline{CS}$, $\overline{WE}$, 74LS93 reset.
- Connections:

Gate 1 (pins 1, 2, 3):
Pin 1 --- 74LS73 Q1 (pin 5)
Pin 2 --- VCC
Pin 3 --- 2114 $\overline{CS}$ (pin 17)
Gate 2 (pins 4, 5, 6):
Pin 4 --- 74LS73 Q1 (pin 5)
Pin 5 --- S11 (read/write)
Pin 6 --- 2114 $\overline{WE}$ (pin 10)
Gate 3 (pins 8, 9, 10):
Pin 8 --- 74LS85 O_A=B (pin 6)
Pin 9 --- 74LS73 Q1 (pin 5)
Pin 10 --- 74LS93 R0(1) (pin 2), R0(2) (pin 3), S13
Gate 4 (pins 11, 12, 13):
Pin 11 --- 74LS73 Q1 (pin 5)
Pin 12 --- S11 (write = 0)
Pin 13 --- 8226 DEN (pin 7)
Pin 14 (VCC) --- VCC & C4 [0.1μF] --- GND
Pin 7 (GND) --- GND

Before 8226

Gate 1 (pins 1, 2, 3):
Pin 1 --- 74LS73 Q1 (pin 5)
Pin 2 --- VCC
Pin 3 --- 2114 $\overline{CS}$ (pin 17)
Gate 2 (pins 4, 5, 6):
Pin 4 --- 74LS73 Q1 (pin 5)
Pin 5 --- S11 (read/write, 10kΩ pull-up/pull-down)
Pin 6 --- 2114 $\overline{WE}$ (pin 10)
Gate 3 (pins 8, 9, 10):
Pin 8 --- 74LS85 O_A=B (pin 6)
Pin 9 --- 74LS73 Q1 (pin 5) or Q2 (pin 12)
Pin 10 --- 74LS93 R0(1) (pin 2), R0(2) (pin 3), S13
Gate 4 (pins 11, 12, 13):
Pin 11 --- S13 (reset, 10kΩ pull-down) or Unconnected
Pin 12 --- VCC or Unconnected

Pin 13 --- 74LS93 R0(1), R0(2) or Unconnected
 Pin 14 (VCC) --- VCC & C4 [0.1μF] --- GND
 Pin 7 (GND) --- GND

- Layout: Place 8226, 74LS73, and 74LS00 centrally, near 2114. Route Q1 to multiple ICs.

4. 2114 RAM (18-pin DIP)

- Purpose: Stores 4-bit number at 6-bit address (A6–A9 grounded).
- Connections:

Pin 1 (A0) --- S1 (10kΩ pull-up/pull-down)
 Pin 2 (A1) --- S2 (same)
 Pin 3 (A2) --- S3
 Pin 4 (A3) --- S4
 Pin 5 (A4) --- S5
 Pin 6 (A5) --- S6
 Pin 7 (A6) --- GND
 Pin 8 (A7) --- GND
 Pin 15 (A8) --- GND
 Pin 16 (A9) --- GND
 Pin 11 (I/O1) --- 8226 D00 (pin 4) & 74LS85 B0 (pin 15) & 74LS244 1A1 (pin 2)
 Pin 12 (I/O2) --- 8226 D01 (pin 6) & 74LS85 B1 (pin 14) & 74LS244 1A2 (pin 4)
 Pin 13 (I/O3) --- 8226 D02 (pin 10) & 74LS85 B2 (pin 13) & 74LS244 1A3 (pin 6)
 Pin 14 (I/O4) --- 8226 D03 (pin 12) & 74LS85 B3 (pin 12) & 74LS244 1A4 (pin 8)
 Pin 10 ($\overline{WE}$) --- 74LS00 pin 6
 Pin 17 ($\overline{CS}$) --- 74LS00 pin 3
 Pin 18 (VCC) --- VCC & C1 [0.1μF] --- GND
 Pin 9 (GND) --- GND

- Layout: Place centrally, with A0–A5 to switches (S1–S6), I/O1–I/O4 to switches, 74LS85, 74LS244.

5. Intel 8226 (16-pin DIP)

- Purpose: Creates data bus, isolation during read.
- Connections:

Pin 3 (DI0) --- S7 (10kΩ pull-up/pull-down) & R8 [330Ω] --- L1 --- GND
 Pin 5 (DI1) --- S8 & R9 [330Ω] --- L2 --- GND
 Pin 11 (DI2) --- S9 & R10 [330Ω] --- L3 --- GND
 Pin 13 (DI3) --- S10 & R11 [330Ω] --- L4 --- GND
 Pin 4 (D00) --- 2114 I/O1 (pin 11) & 74LS85 B0 & 74LS244 1A1
 Pin 6 (D01) --- 2114 I/O2 & 74LS85 B1 & 74LS244 1A2
 Pin 10 (D02) --- 2114 I/O3 & 74LS85 B2 & 74LS244 1A3
 Pin 12 (D03) --- 2114 I/O4 & 74LS85 B3 & 74LS244 1A4
 Pin 7 (DEN) --- 74LS00 pin 13 (write control)
 Pin 9 (DIR) --- S11 (read/write: 0 = write, 1 = read)
 Pin 16 (VCC) --- VCC & C9 [0.1μF] --- GND
 Pin 8 (GND) --- GND
 Pin 1, 2, 14, 15 (NC) --- Unconnected

6. 74LS244 (20-pin DIP, Octal Buffer)

- Purpose: Buffers RAM output for L5–L8 (optional).
- Connections:

Pin 1 ($\overline{1G}$) --- 74LS00 pin 3 ($\overline{CS}$) & S11 (read = 1)
 Pin 2 (1A1) --- 2114 I/O1 (pin 11)
 Pin 4 (1A2) --- 2114 I/O2 (pin 12)

Pin 6 (1A3) --- 2114 I/O3 (pin 13)
 Pin 8 (1A4) --- 2114 I/O4 (pin 14)
 Pin 18 (1Y1) --- R12 [330Ω] --- L5 --- GND
 Pin 16 (1Y2) --- R13 [330Ω] --- L6 --- GND
 Pin 14 (1Y3) --- R14 [330Ω] --- L7 --- GND
 Pin 12 (1Y4) --- R15 [330Ω] --- L8 --- GND
 Pin 19 (2G) --- VCC
 Pin 11–17 (2A1–2A4, 2Y1–2Y4) --- Unconnected
 Pin 20 (VCC) --- VCC & C5 [0.1μF] --- GND
 Pin 10 (GND) --- GND

- Layout: Place near 2114, with outputs to L5–L8 (bottom-right).

7. 74LS85 (16-pin DIP, Comparator)

- Purpose: Compares guess vs. RAM data, drives L9–L11.
- Connections:

Pin 2 (A0) --- S7 & 2114 I/O1
 Pin 3 (A1) --- S8 & 2114 I/O2
 Pin 4 (A2) --- S9 & 2114 I/O3
 Pin 5 (A3) --- S10 & 2114 I/O4
 Pin 15 (B0) --- 2114 I/O1 (pin 11)
 Pin 14 (B1) --- 2114 I/O2 (pin 12)
 Pin 13 (B2) --- 2114 I/O3 (pin 13)
 Pin 12 (B3) --- 2114 I/O4 (pin 14)
 Pin 5 (0_A>B) --- R16 [330Ω] --- L9 (Red) --- GND
 Pin 7 (0_A<B) --- R17 [330Ω] --- L10 (Yellow) --- GND
 Pin 6 (0_A=B) --- R18 [330Ω] --- L11 (Green) --- GND & 74LS00 pin 8
 Pin 10 (I_A>B) --- GND
 Pin 11 (I_A<B) --- GND
 Pin 9 (I_A=B) --- VCC
 Pin 16 (VCC) --- VCC & C6 [0.1μF] --- GND
 Pin 8 (GND) --- GND

- Layout: Place near 2114, with outputs to L9–L11 (right).

8. 74LS93 (14-pin DIP, Counter)

- Purpose: Counts guesses, drives L12–L15.
- Connections:

Pin 14 (CKA) --- 74LS73 Q1 (pin 5)
 Pin 1 (CKB) --- Pin 12 (QA)
 Pin 12 (QA) --- R19 [330Ω] --- L12 --- GND & Pin 1 (CKB)
 Pin 9 (QB) --- R20 [330Ω] --- L13 --- GND
 Pin 8 (QC) --- R21 [330Ω] --- L14 --- GND
 Pin 11 (QD) --- R22 [330Ω] --- L15 --- GND
 Pin 2 (R0(1)) --- 74LS00 pin 10 & S13 (10kΩ pull-down)
 Pin 3 (R0(2)) --- Same as R0(1)
 Pin 5 (VCC) --- VCC & C7 [0.1μF] --- GND
 Pin 10 (GND) --- GND
 Pin 4, 6, 7, 13 (NC) --- Unconnected

- Layout: Place near 74LS85, with outputs to L12–L15 (bottom).

9. Switches (S1–S11)

- Connections:

S1–S6 (A0–A5):
 Each: --- 2114 pins 1–6 & R23–R28 [10kΩ pull-up] --- VCC & R29–R34 [10kΩ pull-down] --- GND
 S7–S10 (D0–D3):

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S7: --- 2114 I/O1 & 74LS85 A0 & 74LS244 1A1 & R8 [330Ω] --- L1 --- GND & R35 [10kΩ pull-up]
--- VCC & R36 [10kΩ pull-down] --- GND
S8: --- 2114 I/O2 & 74LS85 A1 & 74LS244 1A2 & R9 [330Ω] --- L2 --- GND & R37 [10kΩ] --- VCC
& R38 [10kΩ] --- GND
S9: --- 2114 I/O3 & 74LS85 A2 & 74LS244 1A3 & R10 [330Ω] --- L3 --- GND & R39 [10kΩ] ---
VCC & R40 [10kΩ] --- GND
S10: --- 2114 I/O4 & 74LS85 A3 & 74LS244 1A4 & R11 [330Ω] --- L4 --- GND & R41 [10kΩ] ---
VCC & R42 [10kΩ] --- GND
S11 (Read/Write):
--- 74LS00 pin 5 & R43 [10kΩ pull-up] --- VCC & R44 [10kΩ pull-down] --- GND

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10. Push-Buttons (S12, S13)

- Connections:

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S12 (LOAD): --- 74LS73 CLK1 (pin 1) & R45 [10kΩ pull-up] --- VCC & R46
[10kΩ pull-down] --- GND
S13 (Reset): --- 74LS93 R0(1) (pin 2), R0(2) (pin 3) & R47 [10kΩ pull-
down] --- GND

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11. LEDs (L1–L15)

- Connections:

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L1: S7 --- R8 [330Ω] --- Anode --- Cathode --- GND
L2: S8 --- R9 [330Ω] --- Anode --- Cathode --- GND
L3: S9 --- R10 [330Ω] --- Anode --- Cathode --- GND
L4: S10 --- R11 [330Ω] --- Anode --- Cathode --- GND
L5: 74LS244 1Y1 --- R12 [330Ω] --- Anode --- Cathode --- GND
L6: 74LS244 1Y2 --- R13 [330Ω] --- Anode --- Cathode --- GND
L7: 74LS244 1Y3 --- R14 [330Ω] --- Anode --- Cathode --- GND
L8: 74LS244 1Y4 --- R15 [330Ω] --- Anode --- Cathode --- GND
L9: 74LS85 0_A>B --- R16 [330Ω] --- Anode (Red) --- Cathode --- GND
L10: 74LS85 0_A<B --- R17 [330Ω] --- Anode (Yellow) --- Cathode --- GND
L11: 74LS85 0_A=B --- R18 [330Ω] --- Anode (Green) --- Cathode --- GND
L12: 74LS93 QA --- R19 [330Ω] --- Anode --- Cathode --- GND
L13: 74LS93 QB --- R20 [330Ω] --- Anode --- Cathode --- GND
L14: 74LS93 QC --- R21 [330Ω] --- Anode --- Cathode --- GND
L15: 74LS93 QD --- R22 [330Ω] --- Anode --- Cathode --- GND

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12. Power Supply

- Connections:

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VCC: 5V --- All IC VCC pins & switch pull-ups
GND: --- All IC GND pins, LED cathodes, switch pull-downs
C1–C7 [0.1μF]: Between VCC and GND near each IC

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Data Flow with 8226

- Write Operation:
 - S11 = 0 (write), S12 pressed → 74LS73 Q1 pulses.
 - 74LS00:
$$\overline{CS} = 0, \overline{WE} = 0, DEN = 1 \text{ (Q1 NAND S11).}$$
 - 8226: DIR = 0 (S11), DEN = 1, transfers S7–S10 (DI0–DI3) to 2114 I/O1–I/O4 (DO0–DO3).
 - 2114 writes data at address A0–A5 (S1–S6).
- Read Operation:
 - S11 = 1 (read), S12 pressed → Q1 pulses.

- 74LS00:
 $\overline{CS} = 0, \overline{WE} = 1, DEN = 0.$
 - 8226: Disabled ($DEN = 0$), no output to I/O1–I/O4.
 - 2114 outputs data to 74LS85 B0–B3 (comparison) and 74LS244 1A1–1A4 (L5–L8).
 - Gameplay: Unchanged; player sets S7–S10, presses S12, sees L9–L11 (74LS85) and L12–L15 (74LS93).
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Notes

- Improvement: The 8226 eliminates contention by isolating S7–S10 during read, ensuring clean write operations.
 - IC Count: Now 8, but justified for reliability.
 - Control Logic: Fully utilizes 74LS00 (all four gates).
 - Timing: 8226 propagation delay (20 ns) is negligible vs. 2114 (200 ns) and 1 kHz clock (1 ms).
 - Power: ~150–200 mA with additional IC.
 - Alternative: A 74LS245 (octal bus transceiver) could replace the 8226, but it's overkill (8-bit vs. 4-bit needed).
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Visual Layout for Graphical Schematic

- Top-Left: LM555 (clock) with R1, R2, C8.
 - Top-Center: 74LS73, 74LS00, S12 (LOAD), S13 (reset), S11 (read/write).
 - Center: 2114 RAM, with S1–S6 (address), S7–S10 (data), L1–L4 (input LEDs).
 - Right: 74LS85, with L9–L11 (feedback LEDs).
 - Bottom-Right: 74LS244, with L5–L8 (output LEDs).
 - Bottom: 74LS93, with L12–L15 (counter LEDs).
 - Buses:
 - 6-bit address (A0–A5): S1–S6 to 2114.
 - 4-bit data (D0–D3): S7–S10 to 2114, 74LS85, 74LS244, L1–L4.
 - Power: VCC, GND rails across top/bottom, capacitors near ICs.
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